
EDUVISION DV

Higher Education Performance Dashboard

MODULE 8 • DOCUMENTATION & PROJECT DELIVERY

Project	EduVision DV
Platform	Tableau Desktop / Tableau Public (optional)
Dashboard suite	University Overview • Research Analytics • Student Analytics • Country Comparison
Data snapshot	603 university records; 26 columns
Purpose	Interactive higher-education performance analysis
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Module 8 is the final documentation and delivery stage. It consolidates the dataset sources, KPI definitions, dashboard guide, analytics methodology, project structure, and delivery artifacts into a portfolio-ready project.

1. Module 8 Overview

The objective of Module 8 is to make the completed EduVision DV project understandable, reproducible, maintainable, and ready for delivery. The documentation covers the data used, calculated education KPIs, dashboard functionality, analytical methodology, project organization, and final delivery package.

Module 8 objectives

- Document dataset sources and the final analytical dataset.
- Define the major education-performance KPIs used by the dashboard.
- Explain the four interactive dashboards and their visualizations.
- Document filters, navigation, reset behavior, and dashboard interactions.
- Describe the methodology used to compare universities, regions, and countries.
- Organize scripts, data, dashboard files, and documentation into a clear project structure.
- Prepare the project for GitHub delivery and optional Tableau Public deployment.

Module 8 deliverables

Deliverable	Purpose
GitHub Repository	Central location for source code, data references, dashboard files, and documentation.
Final Documentation	Explains the project, data, KPIs, methodology, dashboards, and usage.
Tableau Workbook	Contains the integrated dashboard suite and supporting worksheets.

2. Dataset & Data Preparation

EduVision DV uses a higher-education university dataset prepared for Tableau analysis. The final project dataset contains 603 university records and 26 columns. The dataset combines university identity, location, ranking, research, student, and derived KPI information.

Key dimensions

Dimension	Examples
University	University name / institution identifier
Location	Country, City, Region
Time	Year
Institution type	University Type
Ranking	QS Rank, QS Score, THE Rank, THE Overall
Research	Research Output, Research Impact Score, Research Productivity Index
Students & faculty	International Students, Faculty Count, Student-Faculty Ratio

Derived KPI fields

KPI	Purpose
Global Ranking Score	Overall comparative university performance indicator used throughout the dashboards.
Research Impact Score	Represents research impact for comparative analysis.
Faculty-to-Student Ratio	Shows the relative faculty availability compared with students.
International Student Percentage	Measures the international component of the student population.
Academic Reputation Score	Represents academic reputation for institutional comparison.
Research Productivity Index	Supports comparison of research productivity across institutions/regions.

3. Education Analytics Methodology

The analytical workflow moves from raw higher-education information to cleaned data, derived education KPIs, Tableau worksheets, and integrated dashboards. The purpose is to support comparison rather than simply display individual records.

- 1. Data collection** — Collect university ranking and performance indicators and bring them into a common structure.
- 2. Data cleaning** — Remove duplicates, standardize university/country names, normalize ranking metrics, and prepare Tableau-ready data.
- 3. KPI engineering** — Calculate the project-level performance indicators used for ranking, research, student, and regional analysis.
- 4. Visualization** — Build worksheets using bars, scatter plots, maps, pie/donut-style views, treemaps, tables, box plots, and comparison charts.
- 5. Dashboard integration** — Combine worksheets into four dashboards and connect global filters, navigation, and actions.
- 6. Validation** — Check KPI calculations, ranking calculations, dashboard interactions, and educational metrics.
- 7. Documentation & delivery** — Document the final solution and organize the repository and Tableau workbook for portfolio use.

Analytical principles

- Use aggregated measures when comparing regions and countries.
- Use institution-level views when comparing individual universities.
- Use filters to move from the global dataset to a focused subset.
- Use multiple visualization types so ranking, distribution, relationship, and variation can be viewed separately.
- Use the same core KPI definitions across dashboards to maintain consistency.

4. Dashboard Guide

EduVision DV contains four integrated dashboards: University Overview, Research Analytics, Student Analytics, and Country Comparison. The dashboard navigation allows the user to move between these analytical perspectives.

4.1 University Overview

Visualization	Documentation
Top University Rankings	Horizontal bar chart comparing universities using Global Ranking Score.
Academic Reputation Analysis	Compares universities using Academic Reputation Score.
Global Score vs Research Impact	Scatter plot showing the relationship between Global Ranking Score and Research Impact Score.
Global University Distribution	Map-based geographic view of university locations.
University Performance Trends	Regional comparison of average Global Ranking Score.
Institutional Comparison	Compares selected institutions using average Global Ranking Score.

4.2 Research Analytics

Visualization	Documentation
Publications Analysis	Visual distribution of publication-related data in the selected view.
Citation Performance	Scatter-based comparison of research-related institutional performance.
Research Productivity Trends	Compares research productivity across regions.
Top Research Institutions	Treemap highlighting institutions using block size for visual comparison.
Regional Research Impact	Horizontal bar comparison of average Research Impact Score by region.
Research Impact Comparison	Distribution/comparison view of Research Impact Score.

4.3 Student Analytics

Visualization	Documentation
International Student Analysis	Horizontal bar chart comparing universities by international student percentage.
Student Diversity Trends	Pie chart showing the selected student distribution.
Faculty-to-Student Ratio Analysis	Dot-based comparison of institutions by faculty-to-student ratio.
Student Enrollment Distribution	Regional view of international student enrollment distribution.
Student Distribution Analysis	Treemap-style comparison using student-size categories.
Enrollment Comparisons	Regional comparison of international student enrollment.

4.4 Country Comparison

Visualization	Documentation
Regional Education Performance	Table comparing regions using academic reputation, global ranking, international student, and research productivity.
Top Performing Countries	Horizontal bar chart ranking countries by performance score.

Global Score Distribution	Binned distribution showing the number of universities within global score ranges.
Education Performance Benchmarking	Scatter-based benchmark view with minimum/maximum reference areas.
Regional Score Variation	Box-plot comparison showing score variation by region.
Country Ranking Comparison	Ranked comparison of countries based on performance score.

5. Dashboard Interactivity & User Guide

The dashboards are designed as an interactive analysis suite rather than static charts. The common controls appear across the dashboard pages so users can maintain a consistent workflow while moving between overview, research, student, and country-level analysis.

Feature	How it is used
Year filter	Restricts the analysis to the selected year.
Region filter	Focuses the analysis on one or more geographic regions.
Country filter	Narrows charts and KPIs to selected countries.
University filter	Focuses the dashboard on selected institutions.
Reset	Clears the active filter selections and returns the dashboard to the default view.
Navigation buttons	Move between Overview, Analytics, Students, and Countries.
Interactive selections	Selecting marks in supported views updates related dashboard content.

Recommended user workflow

- Start with University Overview to understand the overall performance picture.
- Use Year, Region, Country, or University filters to focus the analysis.
- Move to Research Analytics to examine publications, research productivity, institutions, and research impact.
- Move to Student Analytics to study internationalization, student distribution, enrollment, and faculty-to-student relationships.
- Use Country Comparison to compare regional and country-level performance.
- Use Reset before starting a new comparison so previous selections do not affect the next analysis.

6. Project Structure & Delivery

The final project should be organized so that another person can understand where the data, scripts, dashboard workbook, and documentation are stored. The required documentation structure is:

/scripts	Data collection, cleaning, KPI-generation, and supporting Python scripts.
/data	Raw, cleaned, and Tableau-ready datasets used by the project.
/dashboard	Final Tableau workbook and related dashboard artifacts.
/docs	Final documentation, methodology, dashboard guide, and supporting project notes.

Suggested repository contents

- README.md — project overview, setup, dashboard screenshots, and usage instructions.
- data/ — dataset versions used by the project.
- scripts/ — Python scripts/notebooks supporting preparation and KPI engineering.
- dashboard/ — final EduVision DV Tableau workbook.
- docs/ — this final documentation and any supporting QA/testing documents.

Delivery checklist

Check	Status to verify before submission
Dataset	Final Tableau-ready dataset is included or clearly referenced.
KPI definitions	All calculated education KPIs are documented.
Dashboard	Four dashboards are present and navigable.
Filters	Year, Region, Country, and University filters are tested.
Reset	Reset behavior is tested on the integrated dashboards.
Workbook	Final Tableau workbook opens correctly.
Documentation	README and final documentation explain how to use the project.

7. Technology Stack

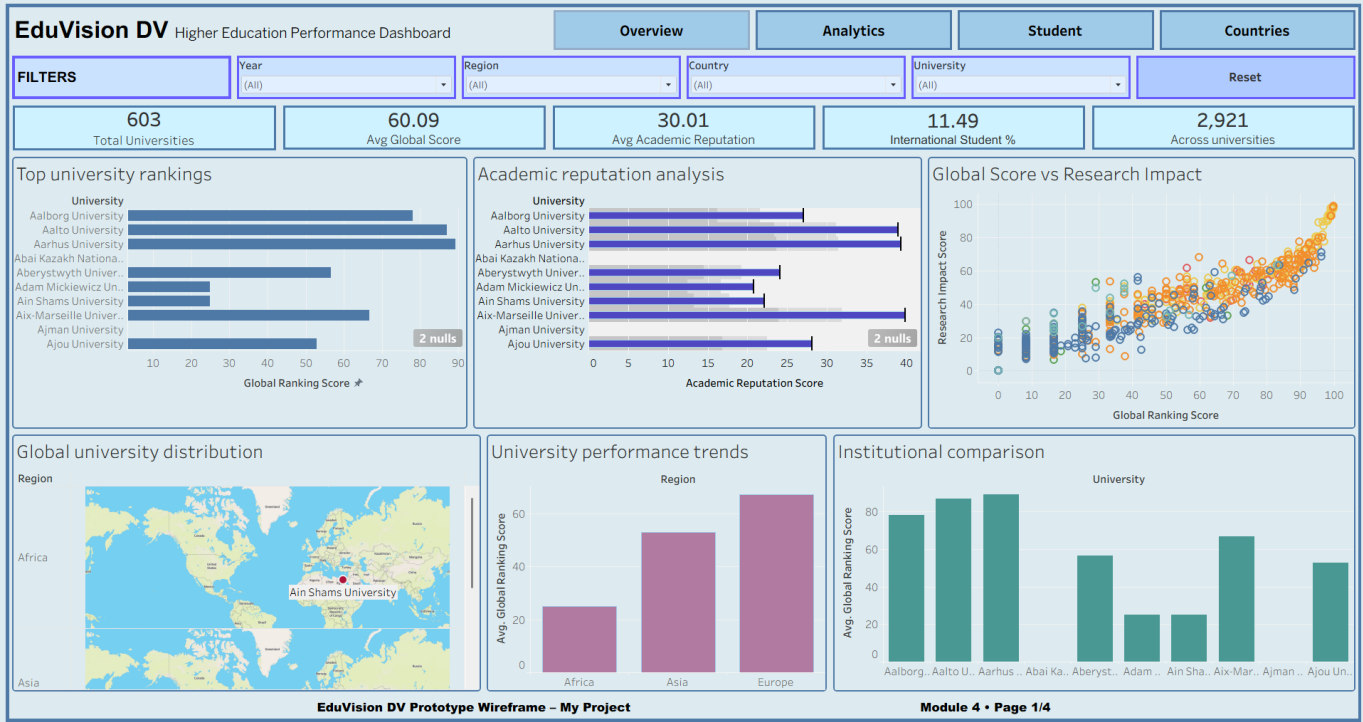
Area	Technology / Tool
Data collection	Python; university ranking datasets
Data processing	Pandas, NumPy
Data cleaning	Python
KPI engineering	Python / calculated fields
Visualization	Tableau Desktop / Tableau Public
Dashboard integration	Tableau Filters, Parameters, Actions
Documentation & delivery	Markdown, GitHub, PDF

8. Portfolio Positioning

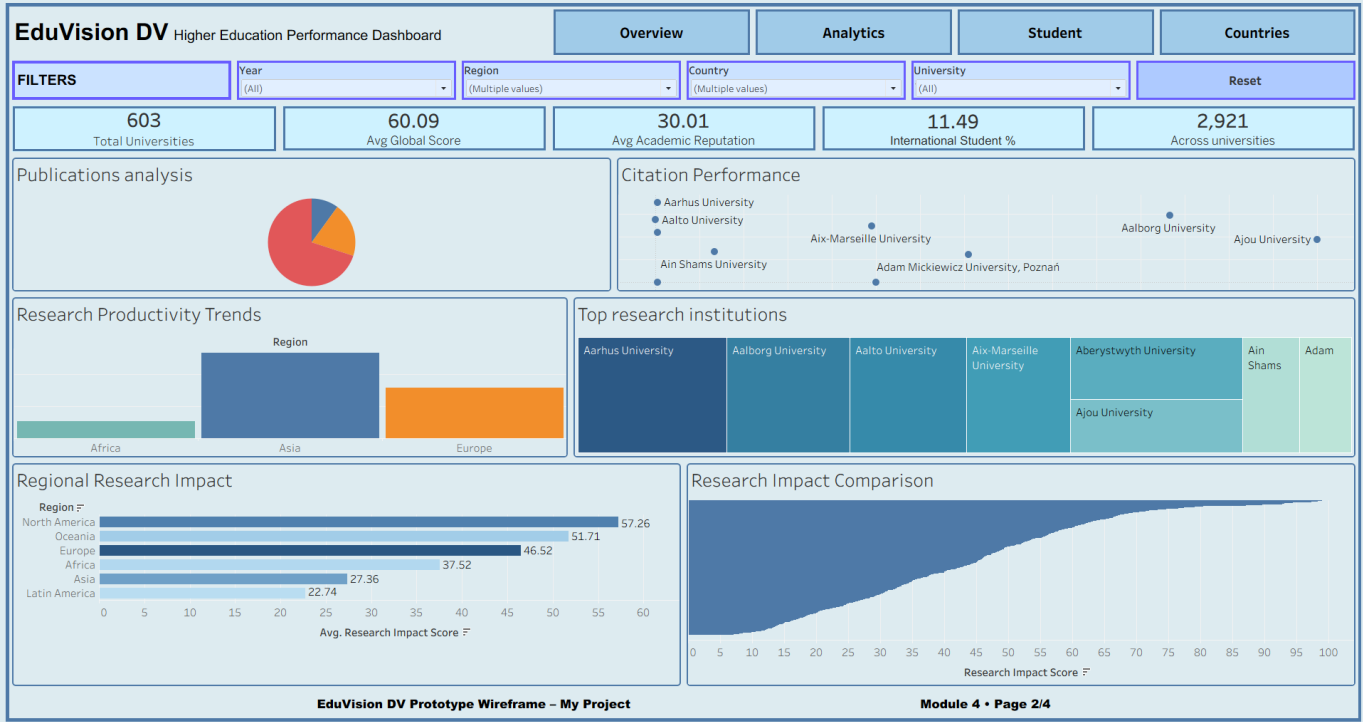
EduVision DV can be presented as an interactive higher-education analytics solution that transforms university-level data into decision-oriented visual analysis. The project demonstrates data preparation, KPI engineering, Tableau visualization, dashboard integration, and documentation in one workflow.

The strongest portfolio message is that the dashboard allows users to move from a high-level university overview to research, student, regional, and country-level comparisons without leaving the same analytical environment.

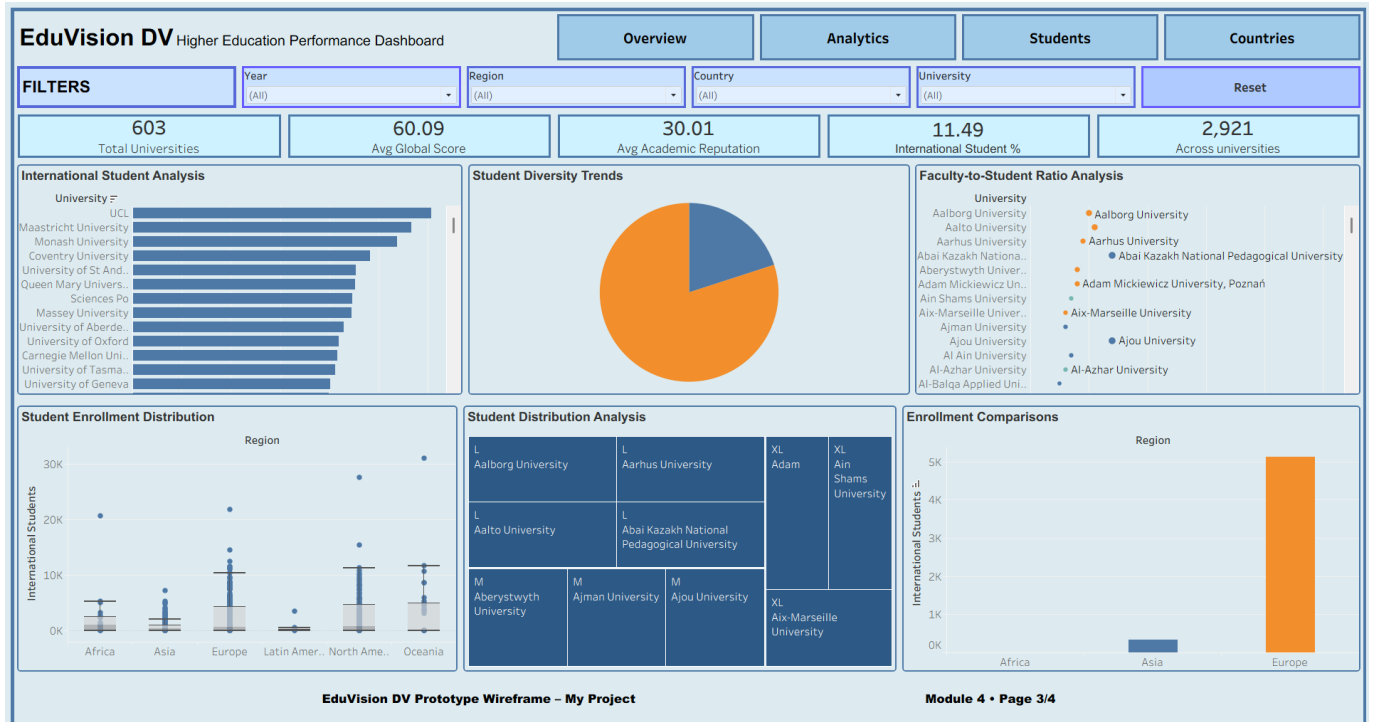
Dashboard evidence 1 — University Overview



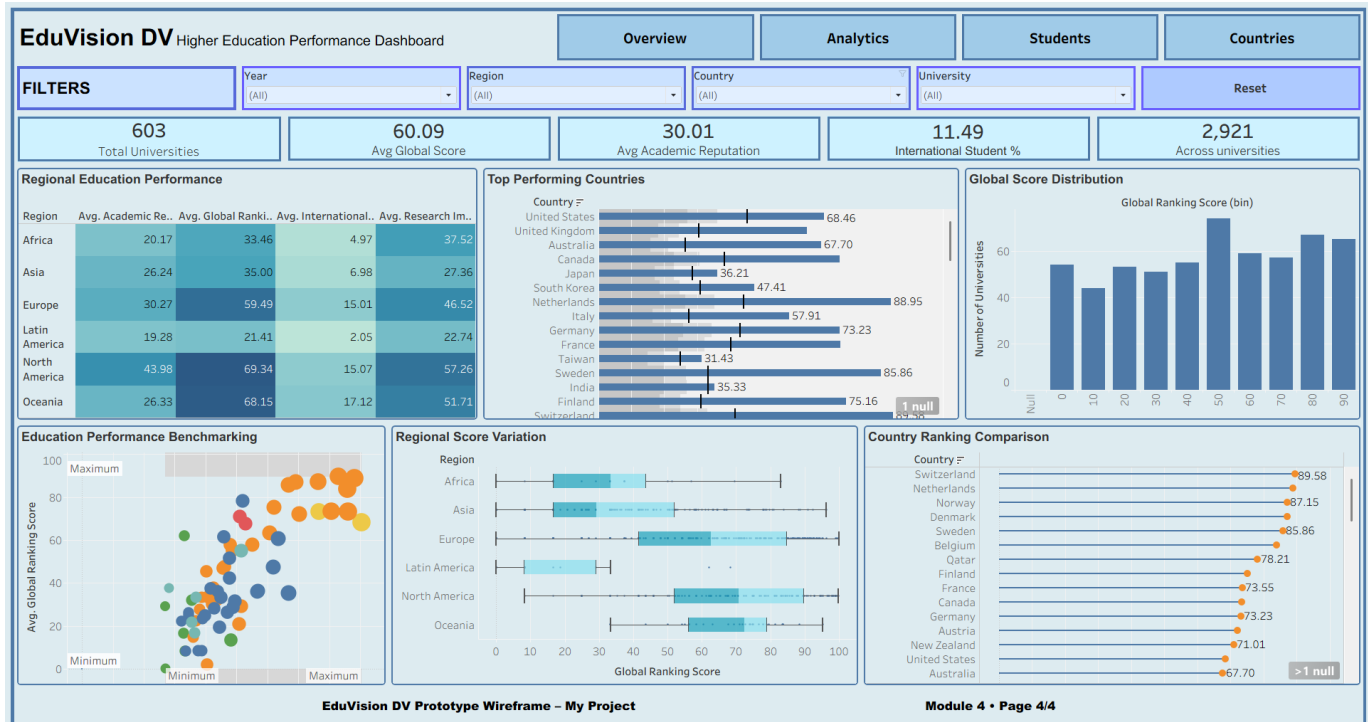
Dashboard evidence 2 — Research Analytics



Dashboard evidence 3 — Student Analytics



Dashboard evidence 4 — Country Comparison



10. Conclusion

Module 8 completes the EduVision DV project by converting the implemented dashboard work into a documented and deliverable solution. The final documentation covers the data foundation, KPI definitions, analytical methodology, four-dashboard guide, interactivity, project structure, technology stack, and delivery checklist.

With the Tableau workbook, organized project repository, and final documentation delivered together, EduVision DV is positioned as a portfolio-ready higher-education performance dashboard suite.

END OF MODULE 8 DOCUMENTATION